

AI-Powered Learning Management & Assessment Platform

PART 1: Teacher-Student Website (MVP)

Purpose: A classroom management and AI-based exam evaluation platform.

Roles

- Teacher
- Student

Teacher Features

- Create classes and generate unique join codes
- Manage students within classes
- Create multiple exams per class
- Add questions and sample answers
- Publish exams and monitor submissions
- AI-based evaluation using sample answers
- Review and override marks manually
- Reveal or hide marks from students
- Generate class, exam, and student reports
- View analytics and performance dashboards

Student Features

- Register and login
- Join classes using class codes
- View announcements and exams
- Submit answers online
- View marks and feedback after publication
- Track progress and performance

Teacher Workflow

Login → Create Class → Generate Code → Students Join → Create Exam → Add Questions & Sample Answers → Publish Exam → Students Submit → AI Evaluation → Review Results → Publish Results → Generate Reports

Student Workflow

Login → Join Class → View Exams → Submit Answers → Wait for Evaluation → View Results & Feedback

Core Database Tables

users, classes, class_members, exams, questions, submissions, answers, announcements, attendance, reports

AI Evaluation Flow

Teacher Sample Answer → Embedding Generation → Student Answer → Similarity Scoring → Marks Generation → Feedback Generation → Store Results

PART 2: Future Administration & Institution Integration

Purpose: Extend the platform from a classroom system into an institution-wide LMS.

New Roles

- Super Admin
- Department Head (Optional)
- Teacher
- Student

Super Admin Features

- Create institutions and departments
- Add and manage teachers
- Monitor all classes and exams
- Institution-wide analytics dashboard
- Department-level reporting
- Teacher performance monitoring
- Student performance monitoring

Department Head Features

- Monitor department classes
- View teacher performance
- Analyze department results
- Generate department reports

Institution Workflow

Admin Login → Create Departments → Add Teachers → Assign Departments → Monitor Classes
→ Monitor Exams → Generate Reports

Additional Database Tables

institutions, departments

users table extended with institution_id and department_id

Institution Analytics

- Total students
- Total teachers
- Total classes
- Total exams
- Department rankings
- Pass percentage analysis

Future Enterprise Features

- AI Question Paper Generator
- OCR-based handwritten answer evaluation
- Plagiarism detection
- AI study recommendations
- Institution leaderboard
- Parent dashboard
- Corporate training support

Recommended Development Roadmap

Phase 1: Teacher-Student Platform (MVP)

Phase 2: Analytics & Reports

Phase 3: OCR & Plagiarism Detection

Phase 4: Institution Administration Module

Phase 5: Department & Multi-Campus Support